

TP Serveur de Virtualisation PowerEdge R730

Introduction

Ce premier travail pratique est consacré à la découverte et à la prise en main du serveur de virtualisation PowerEdge R730. Conçu pour répondre aux exigences des infrastructures informatiques modernes, ce matériel constitue une base d'apprentissage idéale pour les modules d'administration réseaux. Ce document centralise les instructions techniques tout en intégrant en en-tête un rappel des échéances clés de la formation, telles que la remise des documents d'évaluation

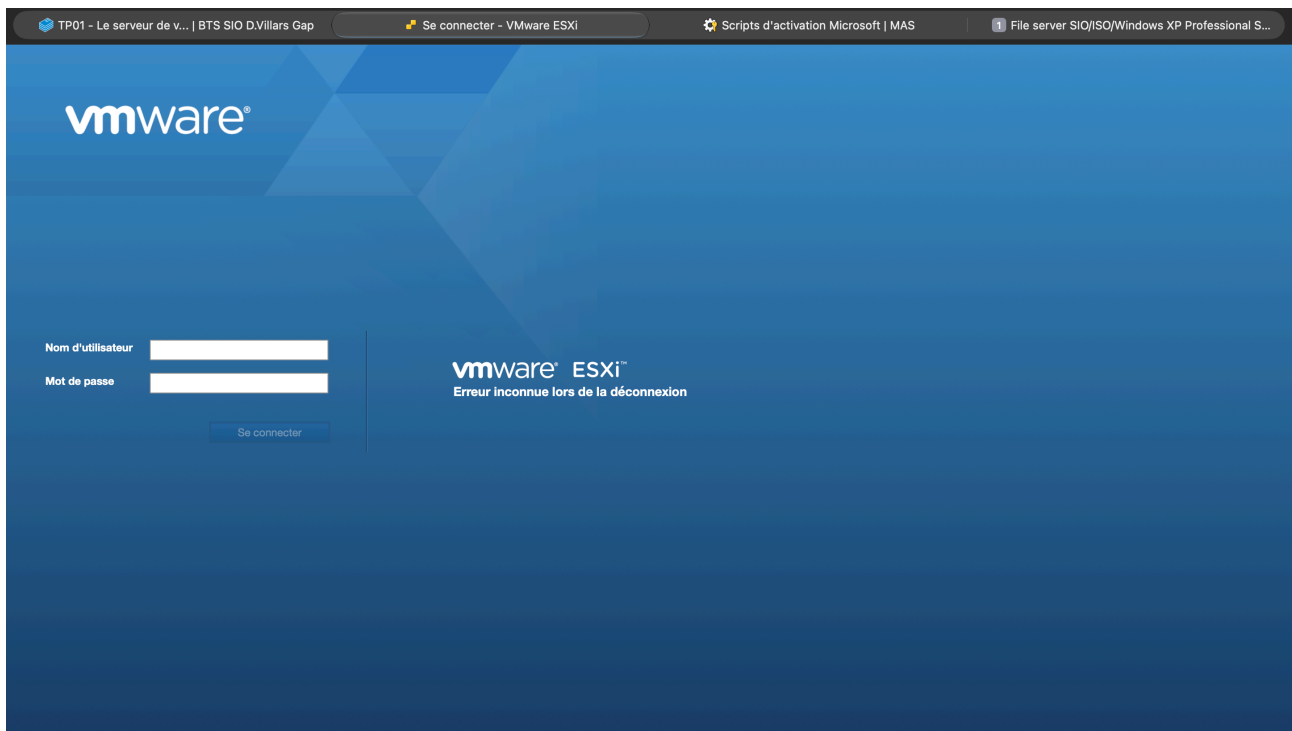
1/ les hyperviseurs existant actuellement sur le marché

Hyperviseur (Type 1 / Bare-Metal)	Éditeur / Projet	Technologie de base	Modèle de Licence
VMware ESXi (vSphere)	Broadcom / VMware	VMkernel (Propriétaire)	Commercial (Abonnement)
Proxmox VE	Proxmox Server Solutions	KVM & LXC (Debian)	Open-source (Gratuit, support payant)
Microsoft Hyper-V	Microsoft	Hyper-V Kernel	Inclus avec Windows Server
XCP-ng	Vates	Xen	Open-source (Gratuit, support payant)
Nutanix AHV	Nutanix	KVM optimisé	Commercial (Bundle HCI Nutanix)
Citrix Hypervisor (XenServer)	Cloud Software Group	Xen	Commercial
Oracle Linux KVM	Oracle	KVM	Gratuit (Support Oracle payant)
Red Hat OpenShift Virtualization	Red Hat	KVM / KubeVirt	Commercial

2/Se connecter au serveur de virtualisation Esxi ayant les caractéristiques suivantes :

Il faut aller sur ce lien <https://172.30.120.101>

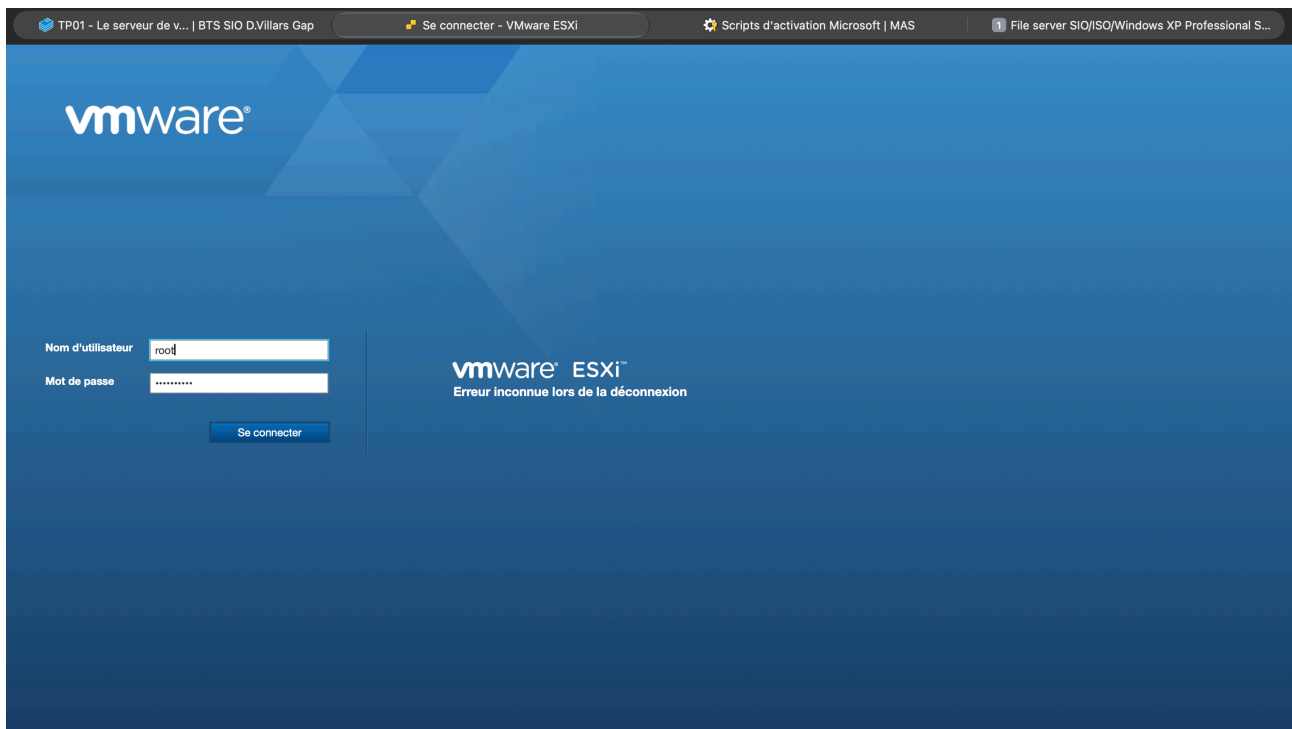
Ce qui donne cette page de connection



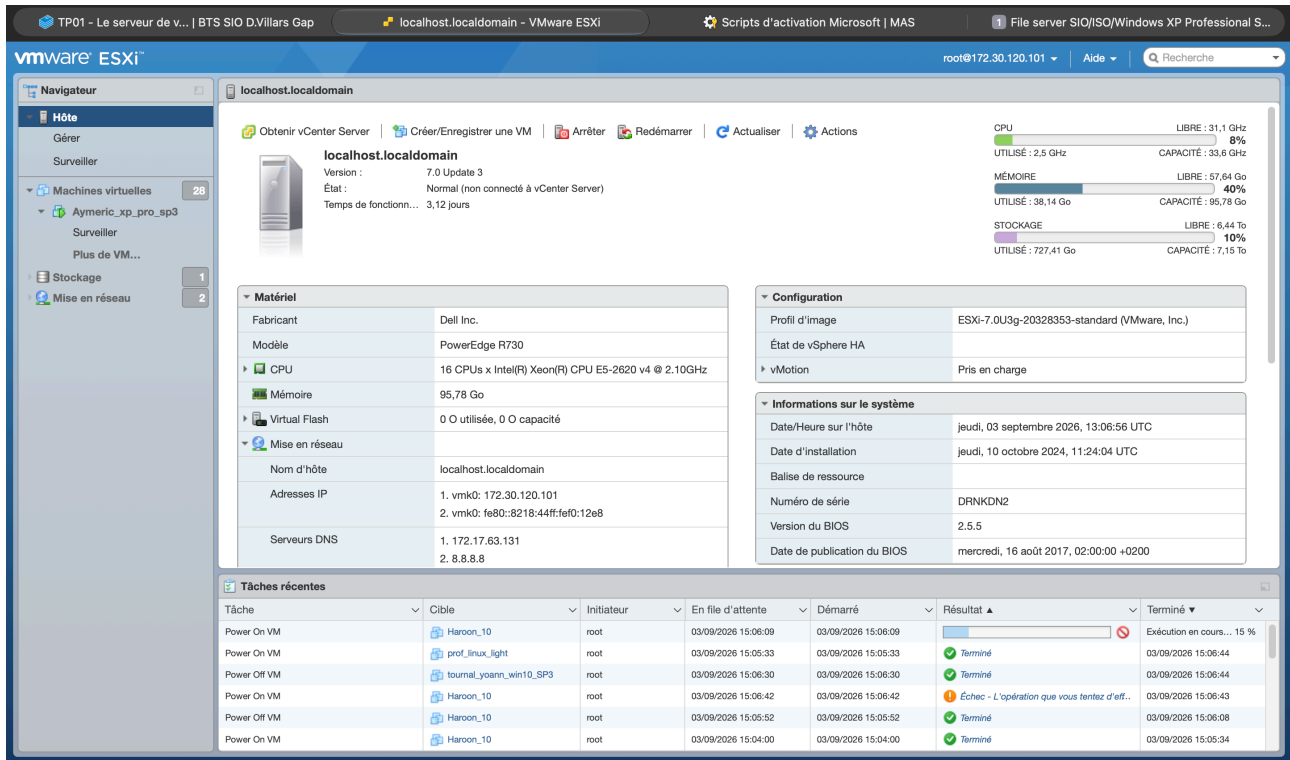
Il faut se connecter avec ce login

User: root

MDP: ?Btssiogap



Pour arriver ici



On peut voir le stockage les information du serveur d'hypervision les réseaux

3/Création de machines virtuelles sur ESXi.

On nous demande de crée cette liste de vm

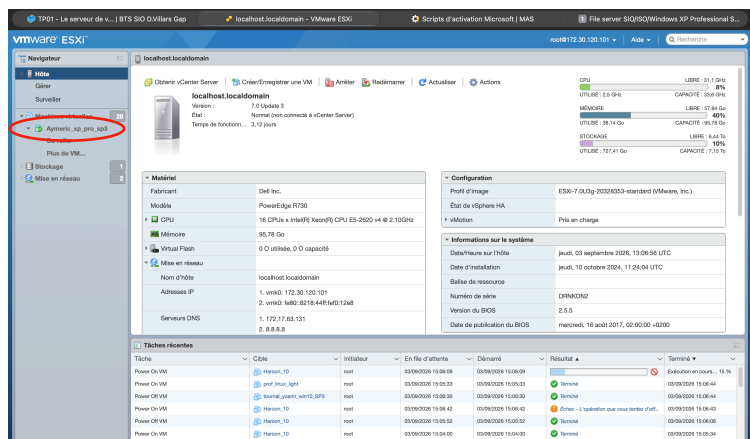
Microsoft Windows

- Windows XP Professionnel (SP3)
- Windows 7 Ultimate (Édition Intégrale)
- Windows 10 Professionnel

Linux (Ubuntu)

- Ubuntu Light (version allégée)
- Ubuntu 26.x

Donc comme on peut le voir sur la capture d'écran j'ai déjà créé ma VM Windows XP



Donc je vais en refaire une autre pour l'exemple

The screenshot shows the VMware ESXi host configuration page for 'localhost.localdomain'. The left sidebar has 'Machines virtuelles' highlighted. The main area displays host details, hardware configuration, system information, and recent tasks.

Host Details:

- Version: 7.0 Update 3
- État: Normal (non connecté à vCenter Server)
- Temps de fonction...: 3,12 jours

Hardware Configuration:

- Fabricant: Dell Inc.
- Modèle: PowerEdge R730
- CPU: 16 CPUs x Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.10GHz
- Mémoire: 95,78 Go
- Virtual Flash: 0 O utilisée, 0 O capacité
- Mise en réseau: Norm d'hôte: localhost.localdomain; Adresses IP: 1. vmk0: 172.30.120.101, 2. vmk0: fe80::8218:44ff:fe0d:12e8; Serveurs DNS: 1. 172.17.63.131, 2. 8.8.8.8

System Information:

- Profil d'image: ESXi-7.0U3g-20328353-standard (VMware, Inc.)
- État de vSphere HA: Pris en charge
- Date/Heure sur l'hôte: jeudi, 03 septembre 2026, 13:06:56 UTC
- Date d'installation: jeudi, 10 octobre 2024, 11:24:04 UTC
- Numéro de série: DRINKDN2
- Version du BIOS: 2.5.5
- Date de publication du BIOS: mercredi, 16 août 2017, 02:00:00 +0200

Recent Tasks:

Tâche	Cible	Initiateur	En file d'attente	Démarré	Résultat	Terminé
Power On VM	Haroon_10	root	03/09/2026 15:06:09	03/09/2026 15:06:09		
Power On VM	prof_linux_light	root	03/09/2026 15:05:33	03/09/2026 15:05:33	Terminé	03/09/2026 15:06:44
Power Off VM	tournal_yoann_win10_SP3	root	03/09/2026 15:06:30	03/09/2026 15:06:30	Terminé	03/09/2026 15:06:44
Power On VM	Haroon_10	root	03/09/2026 15:06:42	03/09/2026 15:06:42	Échec - L'opération que vous tentez d'eff...	03/09/2026 15:06:43
Power Off VM	Haroon_10	root	03/09/2026 15:05:52	03/09/2026 15:05:52	Terminé	03/09/2026 15:06:08
Power On VM	Haroon_10	root	03/09/2026 15:04:00	03/09/2026 15:04:00	Terminé	03/09/2026 15:05:34

On va dans machine virtuelles

The screenshot shows the VMware ESXi Virtual Machines page for 'localhost.localdomain'. The left sidebar has 'Virtual Machines' selected. The main area displays a list of virtual machines with columns for name, status, used space, guest OS, host name, host CPU, and host memory.

Virtual Machines List:

Virtual machine	Status	Used space	Guest OS	Host name	Host CPU	Host memory
Xavier_gros_z6i	Normal	3.08 GB	Microsoft Windows XP Professional (3...	Unknown	39 MHz	1.05 GB
romain_jdt_xp_SP3	Normal	2 GB	Microsoft Windows XP Professional (3...	Unknown	0 MHz	0 MB
chaussegros_xp_SP3	Normal	2 GB	Microsoft Windows XP Professional (3...	Unknown	0 MHz	0 MB
tournal_yoann_win10_SP3	Normal	25 GB	Microsoft Windows XP Professional (3...	Unknown	0 MHz	0 MB
Martins_Nathan_xp_SP3	Normal	12.08 GB	Microsoft Windows XP Professional (3...	Unknown	5 MHz	3.06 GB
Aymeric_xp_pro_sp3	Normal	8.33 GB	Microsoft Windows XP Professional (3...	Unknown	407 MHz	326 MB
raymond_xp_SP3	Normal	15.33 GB	Microsoft Windows XP Professional (3...	Unknown	5 MHz	298 MB
baptiste_xp_SP3	Normal	17.08 GB	Microsoft Windows XP Professional (3...	Unknown	9 MHz	2.06 GB

Recent tasks:

Task	Target	Initiator	Queued	Started	Result	Completed
Power On VM	Aymeric_xp_pro_sp3	root	09/04/2026 09:10:11	09/04/2026 09:10:11	Completed successfully	09/04/2026 09:10:13

Puis on clic sur "create/Register VM"

New virtual machine

- ✓ 1 Select creation type
- 2 Select a name and guest OS
- 3 Select storage
- 4 Customize settings
- 5 Ready to complete

Select creation type

How would you like to create a Virtual Machine?

Create a new virtual machine

Deploy a virtual machine from an OVF or OVA file

Register an existing virtual machine

This option guides you through creating a new virtual machine. You will be able to customize processors, memory, network connections, and storage. You will need to install a guest operating system after creation.

Back Next Finish Cancel

Puis next car on vas pas importer de machine virtuelle

Puis on renseigne le nom de la machine et on sélection la famille et la version

New virtual machine

- ✓ 1 Select creation type
- 2 Select a name and guest OS
- 3 Select storage
- 4 Customize settings
- 5 Ready to complete

Select a name and guest OS

Specify a unique name and OS

Name

Enter a name for the virtual machine.

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Compatibility ESXi 7.0 U2 virtual machine

Guest OS family Select Guest OS Family

Guest OS version Select Guest OS Version

Back Next Finish Cancel

ce qui donne ça pour moi

New virtual machine - Aymeric_win_7 (ESXi 7.0 U2 virtual machine)

✓ 1 Select creation type
2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete

Select a Guest OS Family.

Specify a unique name and OS

Name
Aymeric_win_7

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Compatibility: ESXi 7.0 U2 virtual machine

Guest OS family: Windows

Guest OS version: Microsoft Windows 7 (32-bit)

Back Next Finish Cancel

Puis on sélection le l'endroit de stockage pour moi j'ai que datastore1 puis on clic sur next

New virtual machine - Aymeric_win_7 (ESXi 7.0 U2 virtual machine)

✓ 1 Select creation type
✓ 2 Select a name and guest OS
3 Select storage
4 Customize settings
5 Ready to complete

Select storage

Select the storage type and datastore

Standard Persistent Memory

Select a datastore for the virtual machine's configuration files and all of its virtual disks.

Name	Capacity	Free	Type	Thin pro...	Access
datastore1	7.15 TB	6.44 TB	VMFS6	Supported	Single

1 items

Back Next Finish Cancel

On renseigne la quantité de stockage et de ram que la vm peut utiliser au maximum

Et on sélection datastore ISO puis on clic sur browse

New virtual machine - Aymeric_win_7 (ESXi 7.0 U2 virtual machine)

✓ 1 Select creation type
✓ 2 Select a name and guest OS
✓ 3 Select storage
4 Customize settings
5 Ready to complete

Please select an ISO image for the CDROM

Configure the virtual machine hardware and virtual machine additional options

Hard disk 1	24	GB	
SCSI Controller 0	LSI Logic SAS		
SATA Controller 0			
USB controller 1	USB 2.0		
Network Adapter 1	VM Network	☒ Connect	
CD/DVD Drive 1	Datastore ISO file	☒ Connect	
Status	☒ Connect at power on		
CD/DVD Media			Browse...
Controller location	SATA controller 0	SATA (0:0)	
Video Card	Default settings		

Back Next Finish Cancel

Datastore browser

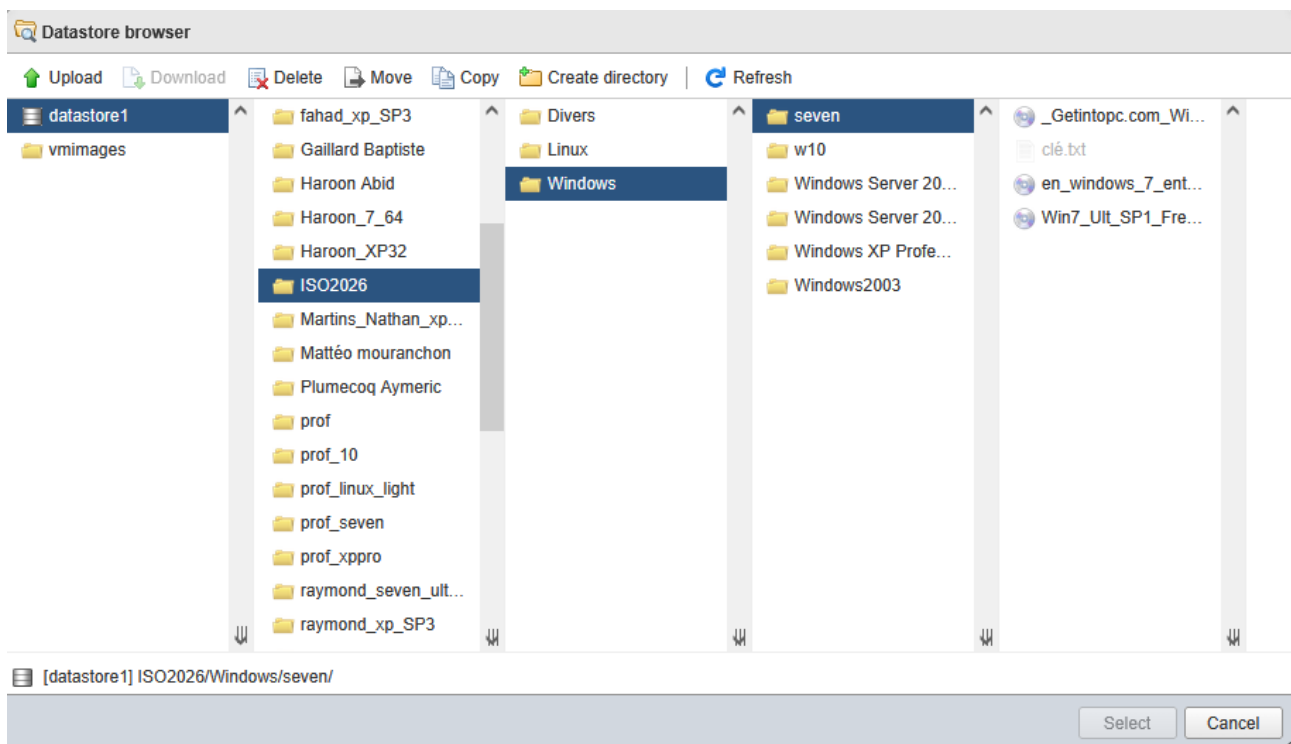
Upload Download Delete Move Copy Create directory Refresh

datastore1
vmimages

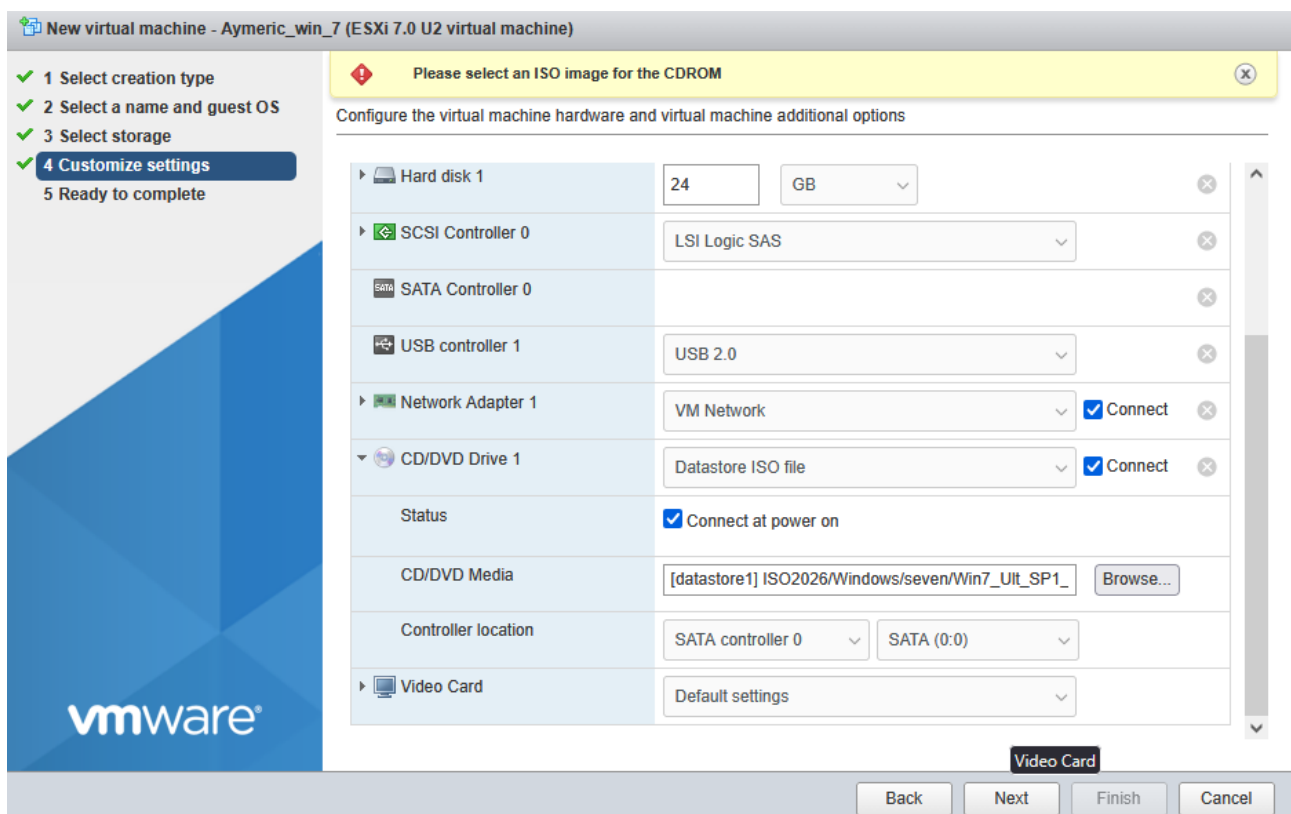
fahad_xp_SP3
Gaillard Baptiste
Haroon Abid
Haroon_7_64
Haroon_XP32
ISO2026
Martins_Nathan_xp...
Mattéo mouranchon
Plumecoq Aymeric
prof
prof_10
prof_linux_light

[datastore1]

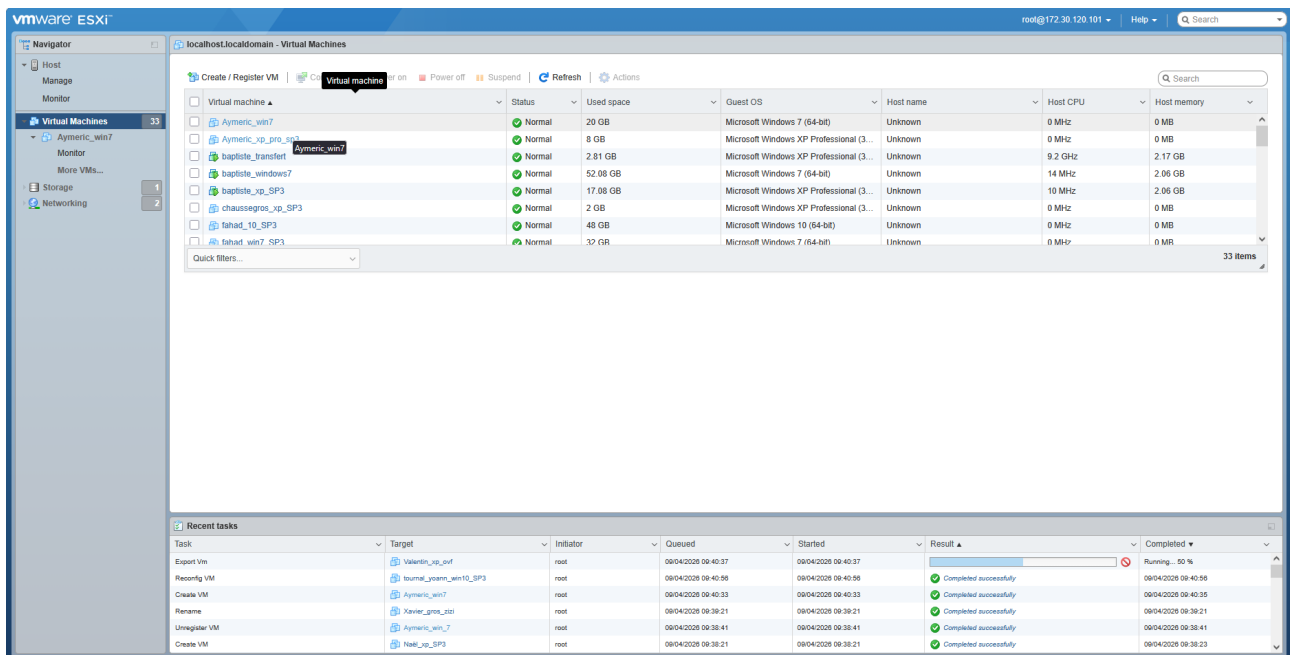
Select Cancel



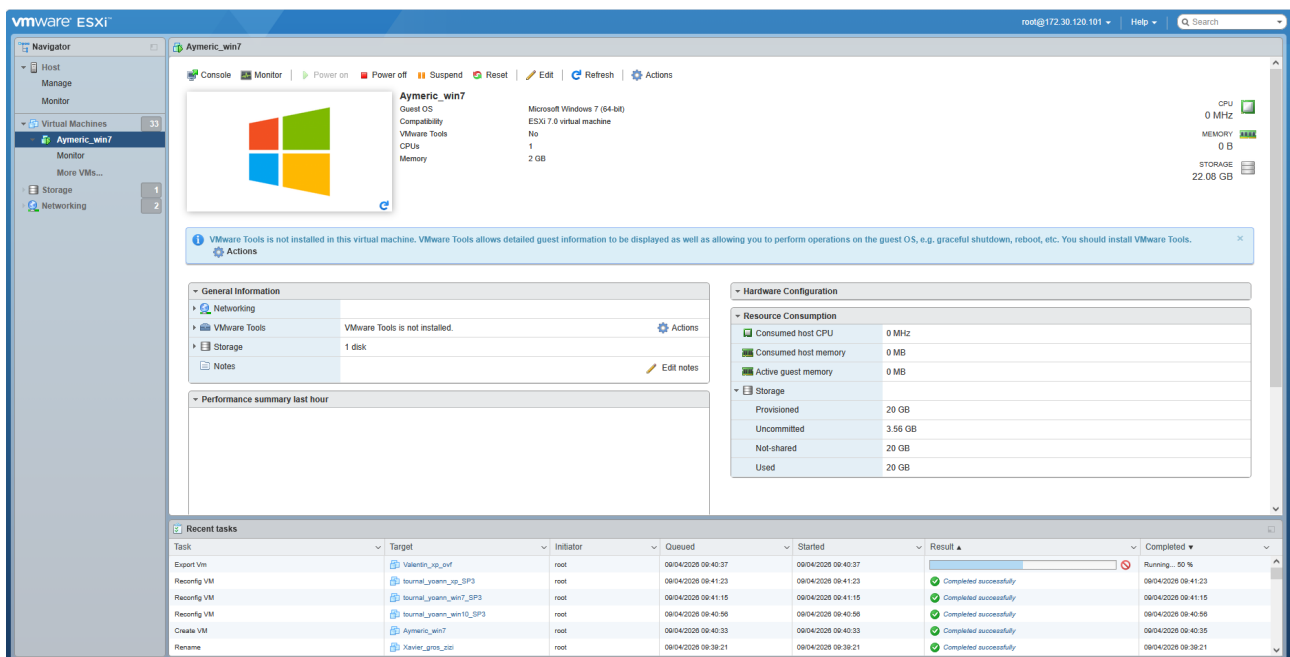
Puis une fois que vous avez sélectionné votre ".iso" ça va donner ça



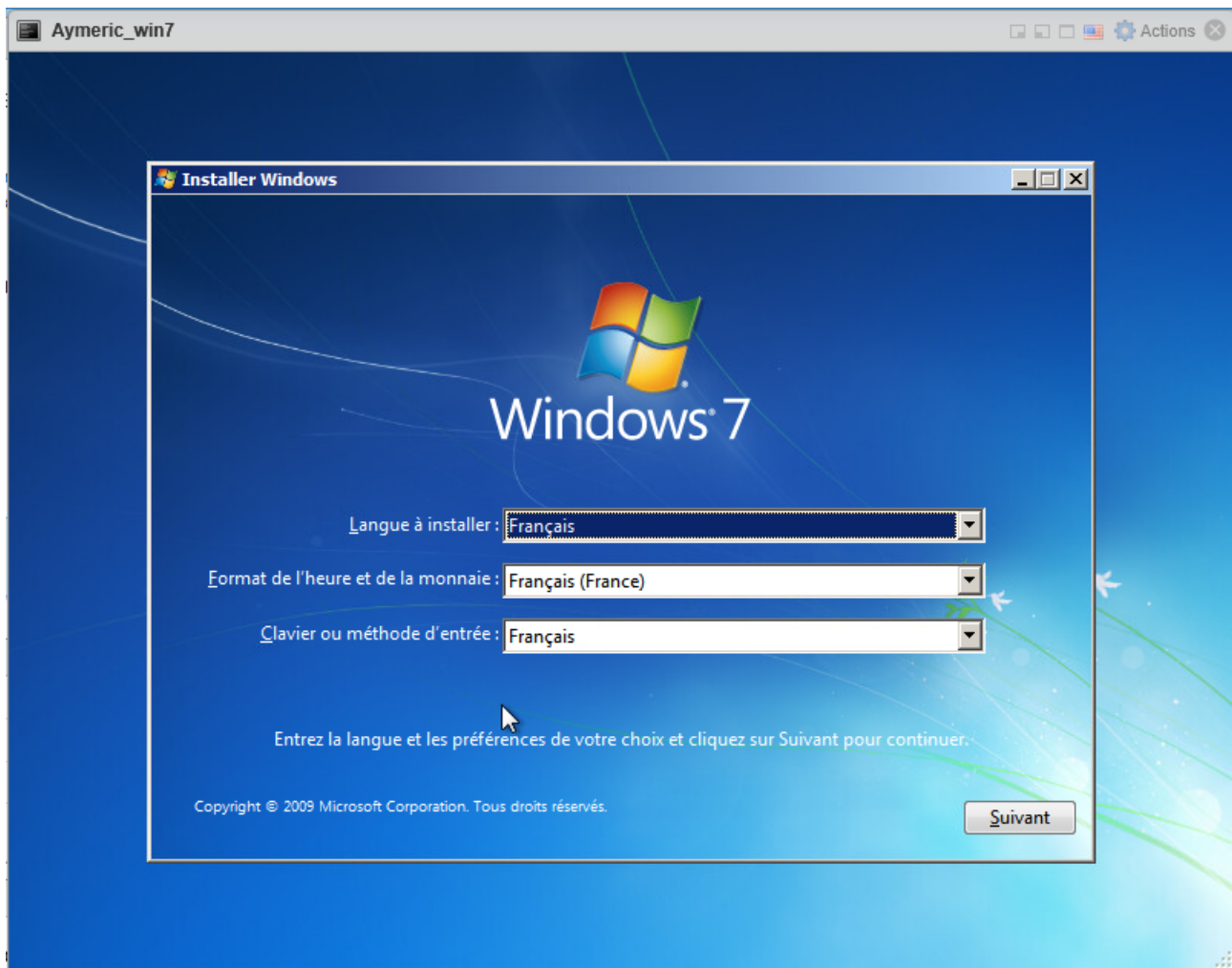
Puis vous faites next puis finish ce qui donne l'image dans de sous



Comme on peut le voir on "Aymeric_win7" est apparu, puis on double clic sur notre vm dans mon cas c'est "Aymeric_win7"



Puis après on double clic sur l'icône windows et la vm vas ce lancer toutes seules voilà il vous reste plus cas faire l'installation

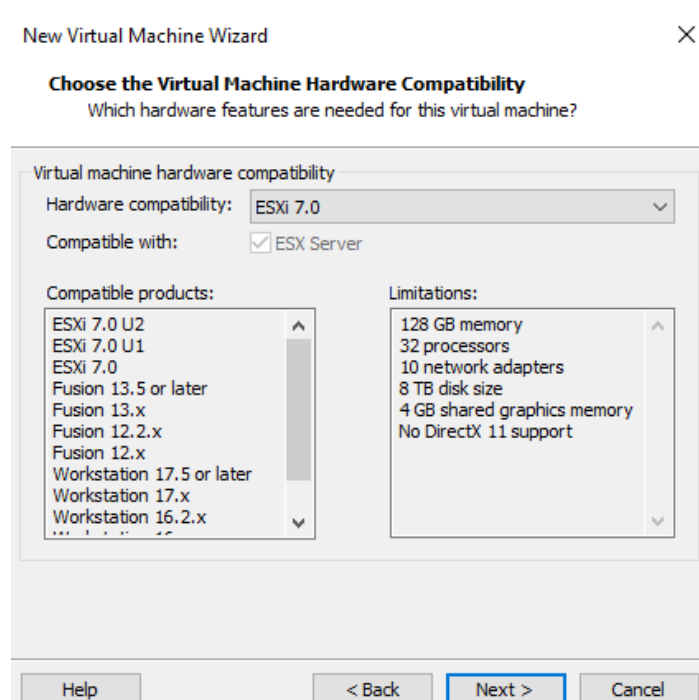


4/importation de la workstation vers le serveur vmware

Il faut faire une vm sur VMware sauf qu'il faut sélection "Custom"

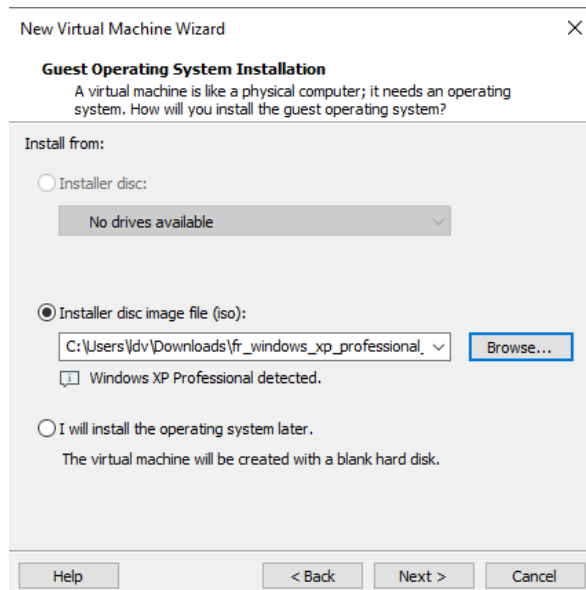


On prend la compatibilité ESXI 7.0



Puis on fait comme d'habitude

Sélection du ".iso"



New Virtual Machine Wizard

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

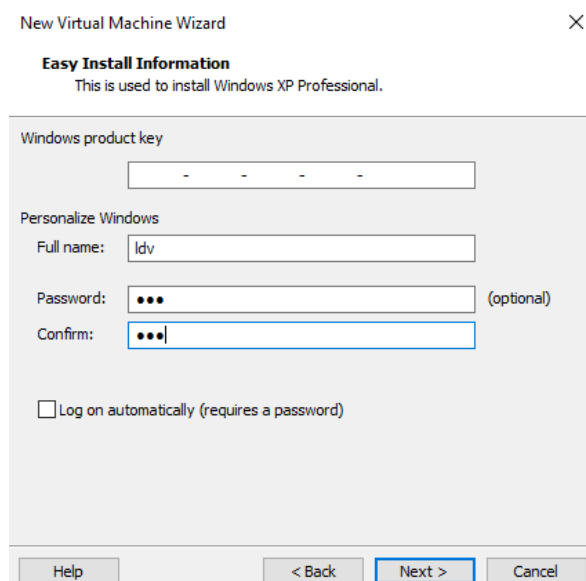
☐ Installer disc:
No drives available

☒ Installer disc image file (iso):
C:\Users\ldv\Downloads\fr_windows_xp_professional_... [Browse...](#)
☒ Windows XP Professional detected.

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

Puis on renseigne clé licence, user, MDP



New Virtual Machine Wizard

Easy Install Information
This is used to install Windows XP Professional.

Windows product key
- - - -

Personalize Windows

Full name: ldv

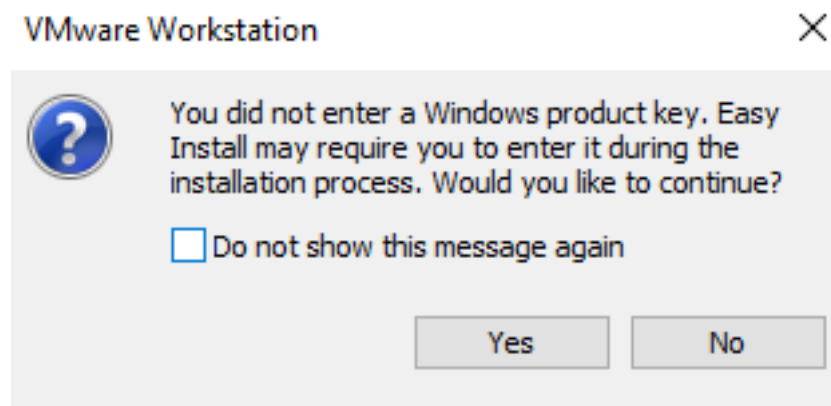
Password: ... (optional)

Confirm: ...

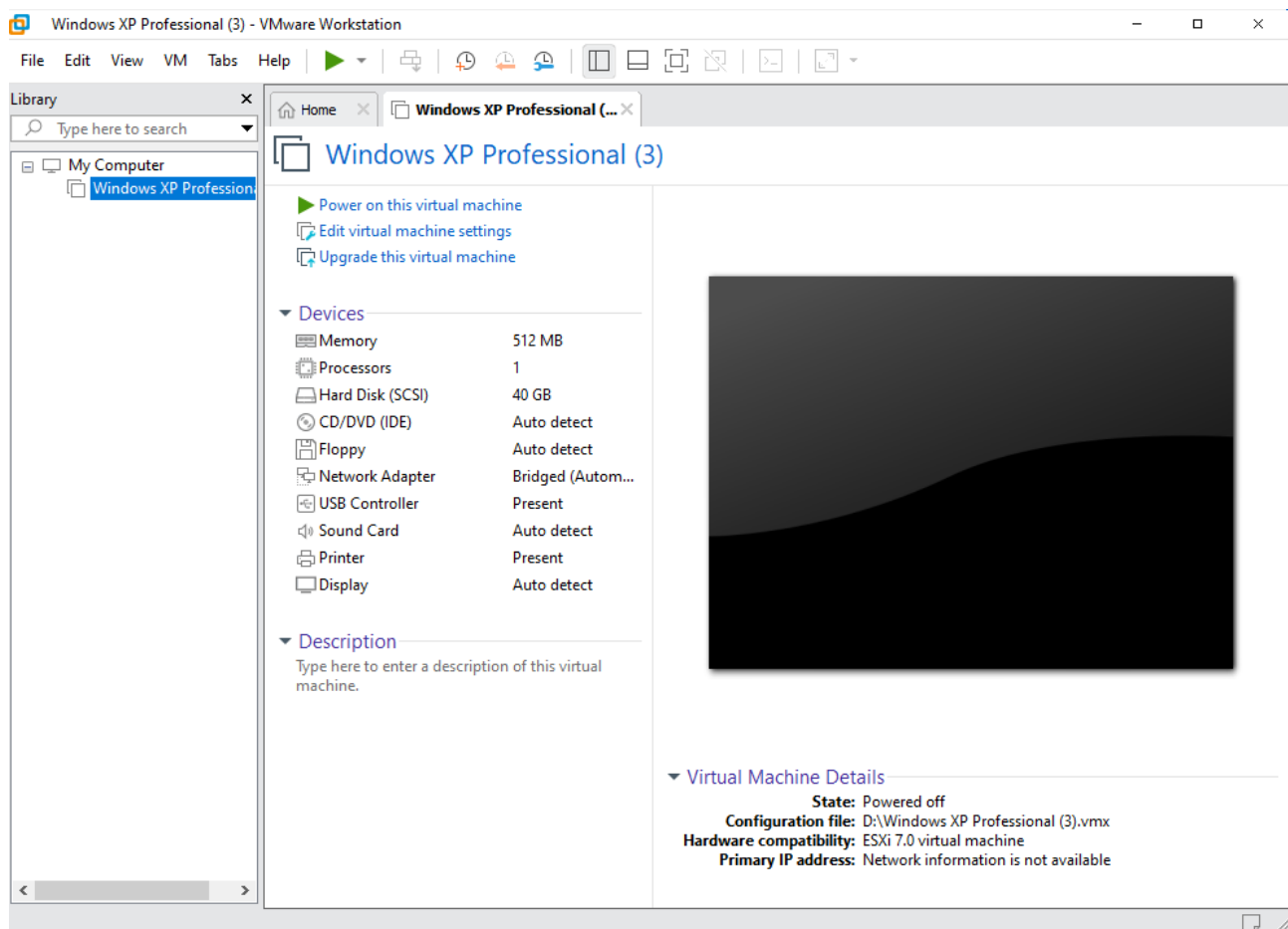
☐ Log on automatically (requires a password)

Help < Back Next > Cancel

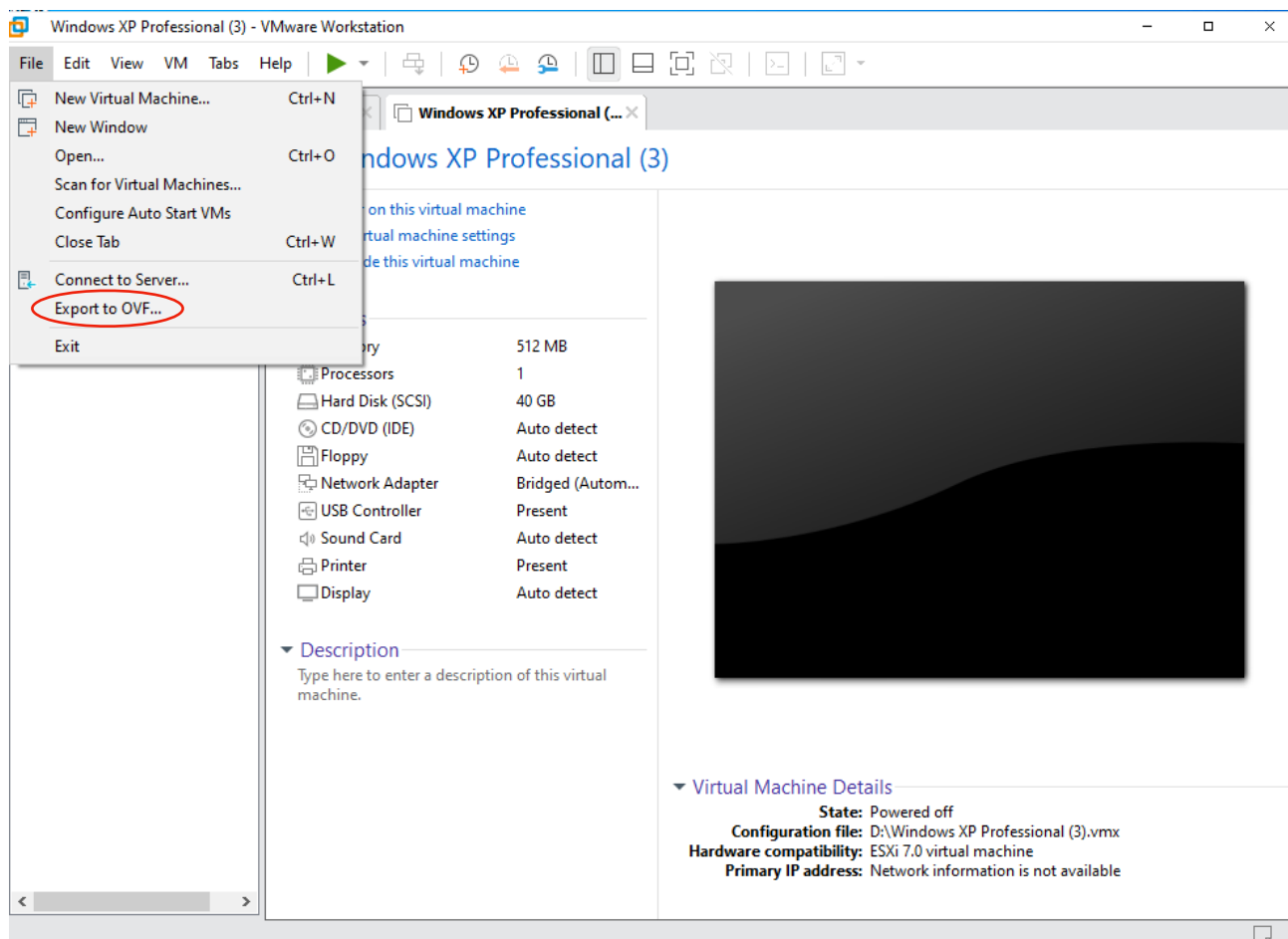
Si jamais il veut pas prendre la clé de licence



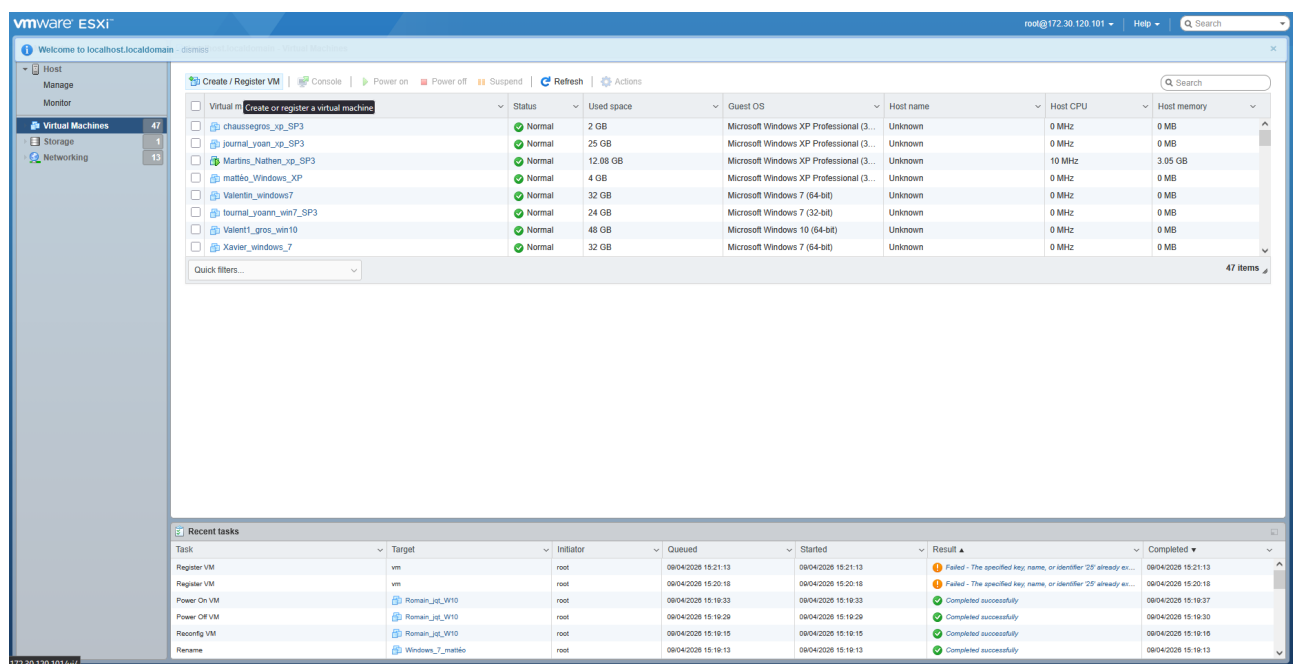
Puis il demande le information hardware et la localisation des fichier puis on a ce résultat



Puis pour exporter vas dans "file" puis export OVF



Puis on fait on décide on le met, puis on retourne sur le serveur et on clic sur carte machine



Puis on sélection l'option en bleu

The screenshot shows the 'New virtual machine' wizard in VMware Workstation. The left sidebar contains a list of steps: 1 Select creation type (highlighted with a green checkmark), 2 Select OVF and VMDK files, 3 Select storage, 4 License agreements, 5 Deployment options, 6 Additional settings, and 7 Ready to complete. The main area is titled 'Select creation type' with the subtitle 'How would you like to create a Virtual Machine?'. It contains three options: 'Create a new virtual machine', 'Deploy a virtual machine from an OVF or OVA file' (highlighted in blue), and 'Register an existing virtual machine'. To the right of these options, a text box explains: 'This option guides you through the process of creating a virtual machine from an OVF and VMDK files.' At the bottom right, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

Puis on renseigne le nom de la machine il faut le fichier OVF et MVK

The screenshot shows the second step of the 'New virtual machine' wizard, titled 'Select OVF and VMDK files'. The left sidebar now highlights step 2. The main area has the subtitle 'Select the OVF and VMDK files or OVA for the VM you would like to deploy'. It includes a text input field for 'Enter a name for the virtual machine.' and a note: 'Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.' Below this is a large light blue rectangular area with the text 'Click to select files or drag/drop'. At the bottom right, the same four buttons ('Back', 'Next', 'Finish', 'Cancel') are present.

New virtual machine - aymeric_xp_transfert

1 Select creation type

2 Select OVF and VMDK files

3 Select storage

4 License agreements

5 Deployment options

6 Additional settings

7 Ready to complete

Select OVF and VMDK files

Select the OVF and VMDK files or OVA for the VM you would like to deploy

Enter a name for the virtual machine.

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

×  Windows XP Professional (3).ovf

×  Windows_XP_Professional_(3)-disk1.vmdk

Back

Next

Finish

Cancel

On fait next jusqu'à cette étape ou on choisi le network où on met VM network

New virtual machine - aymeric_xp_transfert

1 Select creation type

2 Select OVF and VMDK files

3 Select storage

4 Deployment options

5 Ready to complete

Deployment options

Select deployment options

Network mappings	bridged	VM Network
Disk provisioning	☒ Thin ☐ Thick	
Power on automatically	☒	

Back

Next

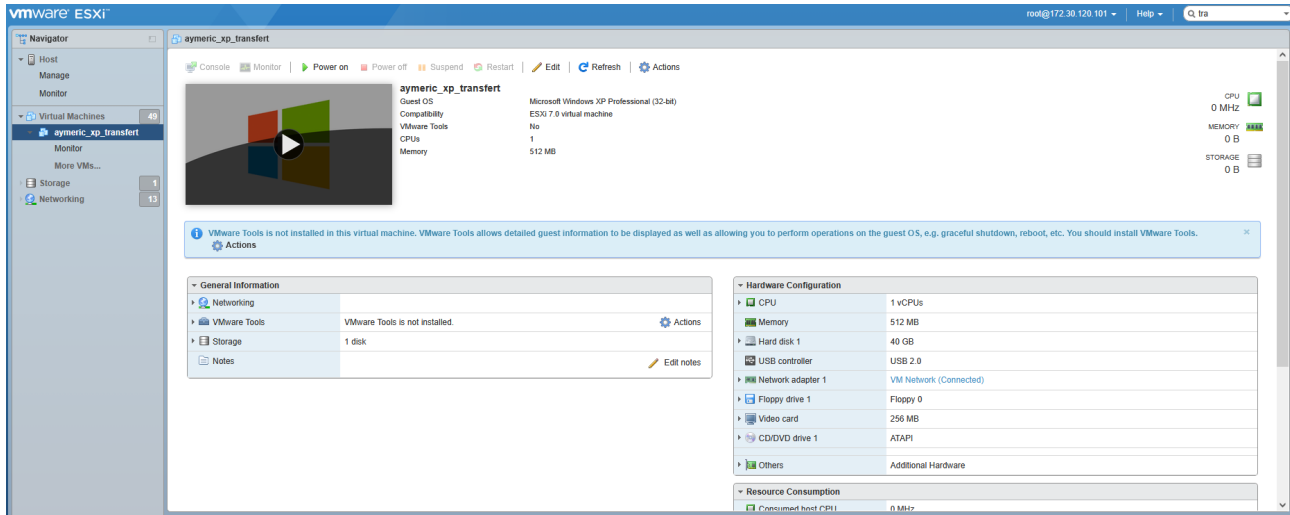
Finish

Cancel

Puis on peut voir ça tout en bas de l'interface du serveur

Recent tasks						
Task	Target	Initiator	Queued	Started	Result	Completed
Upload disk - Windows_XP_Professional_32-bit.vmdk (1 of 1)	aymeric_xp_transfert	root	08/04/2020 15:26:29	08/04/2020 15:26:29		Running... 0 %

On voit qu'elle s'import, puis on voit qu'elle est apparue



Qu'on peut lancer

Conclusion

Voilà comment marche la version serveur de vm